TATA STEEL MAINTAINAI

Industrial Predictive Maintenance & Asset Intelligence Platform | Executive Maintenance Intelligence Briefing

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LEGAL NOTICE: *DISCLAIMER: This application is an independent portfolio demonstration project. It is NOT an official Tata Steel system, does not access proprietary machinery data or internal SCADA, and does not bypass physical safety interlocks. All model predictions are statistical risk estimates, not mechanical certainty. Safety alarms and qualified human inspections always take legal precedence.*

1. EXECUTIVE FLEET HEALTH SUMMARY

Monitored Assets	Critical / High Risk	Active Anomalies	Total Failure Exposure	Est. Net Savings	Modeled ROI
12	2 (0 Crit, 2 High)	0	$525,439	$367,143	231.9%

2. HIGH-PRIORITY MACHINERY ASSET AUDIT

Machine ID	Equipment Type	Production Area	Failure Prob	Risk Score	Category	Expected Loss
BOF-ID-FAN-001	Industrial fans and bl	Steel Melt Shop / BOF	76.8%	63.4	High Risk	$234,215
BF-PUMP-001	Blast furnace auxiliar	Blast Furnace (Ironmak	57.8%	53.9	High Risk	$144,270
RM-MOTOR-002	Industrial motors	Hot Strip Mill (HSM)	42.7%	46.4	Medium Risk	$141,555
BF-BLWR-001	Blast furnace auxiliar	Blast Furnace (Ironmak	0.0%	25.0	Low Risk	$0
LADLE-CRANE-001	Material-handling equi	Steel Melt Shop / BOF	0.0%	25.0	Low Risk	$0
CONT-CAST-001	Steel-processing machi	Continuous Casting Pla	0.0%	25.0	Low Risk	$0

3. PRIORITIZED PREVENTIVE MAINTENANCE RECOMMENDATIONS

Priority	Machine ID	Action Summary & Rationale	Target Window	Est. Cost	Status
Critical	BOF-ID-FAN-001	**Action:** Conduct dynamic balancing of impeller fan assembly; check casing wear plates and... **Reason:** Vibration anomaly detected (Elevated vibration: 9.383 mm/s exceeds operational w	Next Available Planned Shutdow	$10,850	Pending
High	BF-PUMP-001	**Action:** Schedule comprehensive non-destructive predictive inspection (ultrasonic acousti... **Reason:** Multivariate ML model flagged high failure likelihood (57.8%) within 72h.	Next Available Planned Shutdow	$9,975	Pending
Medium	RM-MOTOR-002	**Action:** Perform dynamic vibration FFT analysis; inspect drive-end bearings, foundation a... **Reason:** Vibration anomaly detected (Elevated vibration: 6.852 mm/s exceeds operational w	Next Regular Scheduled Mainten	$12,600	Pending
Routine	HYD-PRESS-001	**Action:** Maintain routine visual inspection and scheduled periodic lubrication interval.... **Reason:** Telemetry operates within nominal baseline reference parameters.	Next Regular Scheduled Mainten	$7,350	Pending

4. BUSINESS IMPACT & FINANCIAL SCENARIO COMPARISON

Scenario Metric	Scenario A (Run-to-Failure)	Scenario B (Planned PM)	Net Variance / Savings
Total Financial Exposure	$525,439.03	$158,295.93	**$367,143.10**
Fleet Availability Projection	99.99%	99.99%	**+0.00%**
Intervention Effectiveness Assumption	0.0% (No Intervention)	90% Risk Mitigation	Target Parameter

Summary Conclusion: By shifting from reactive breakdown remediation to condition-based proactive maintenance synchronized with planned plant shutdowns, the operation preserves high capital equipment uptime while reducing emergency repair premiums and catastrophic molten steel production outages.